1. Competency #3: Implement a given strategy to determine a mathematical solution

(Highly relevant)

- Although no formula is explicitly provided, the problem implies a standard two-stage strategy:
  - First, reverse-engineer the number of DVDs rented by subtracting the membership fee and dividing by the member rate;
  - Then, apply that quantity to the non-member rate.
- Students must recognize and execute this sequence—demonstrating the ability to implement a known procedural approach.
- This is a classic example of applying a structured mathematical method to solve a real-world problem.

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2. Competency #7: Manipulate numbers, data, and information

(Highly relevant)

- Involves multiple numerical operations with decimals:
  - Subtraction: 52.50 − 10 = 42.50
  - Division: 42.50 ÷ 2.50 = 17
  - Multiplication: 3.20 × 17 = 54.40
- All values are grounded in a realistic pricing context (currency, fees), requiring accurate handling of decimal arithmetic.
- Core competence lies in effectively processing quantitative information from everyday situations.

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3. Competency #12: Carry out multi-step procedures

(Highly relevant)

- The solution follows a clear, logical sequence of three computational steps:
  1. Subtract the membership fee from total spending;
  2. Divide the remaining amount by the member rental rate to find the number of DVDs;
  3. Multiply that number by the non-member rate to compute the alternative total cost.
- Each step depends on the result of the previous one, forming a coherent multi-stage procedure.

4. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

(Moderately relevant)

- The context clearly distinguishes between a one-time membership fee and per-item rental fees.
- Misinterpreting the total as only rental costs (and forgetting to subtract the 10 zeds) would lead to error.
- Familiarity with how membership models work in real life (fixed annual fee + discounted usage) helps students correctly decompose the total cost.

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5. Competency #1: Perform a simple calculation

(Moderately relevant)

- All operations are basic arithmetic, but involve decimals (e.g., 42.50 ÷ 2.50), requiring care and fluency.
- This competency provides essential support for higher-level reasoning but is not the primary focus.

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6. Competency #2: Select a strategy to solve a problem

(Weakly relevant)

- While the strategy isn't spelled out, there is essentially only one viable approach: recover the number of rentals first, then recompute under the non-member rate.
- Students aren't required to choose between alternative methods (e.g., algebraic vs. arithmetic), so strategic selection is minimal.

1. 能力 #3：实施给定策略以确定数学解

（高度相关）

- 题目未直接给出公式，但隐含标准策略：
  - 先从会员总支出中反推租片数量（需减去会员费后除以会员单价）；
  - 再用该数量乘以非会员单价。
- 学生必须识别并执行这一两阶段策略，体现了对“逆向求解 + 正向应用”流程的掌握。
- 这是典型的“实施既定数学程序”行为。

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2. 能力 #7：操作数字、数据和信息

（高度相关）

- 涉及多项数值操作：
  - 减法：52.50 − 10 = 42.50
  - 除法：42.50 ÷ 2.50 = 17
  - 乘法：3.20 × 17 = 54.40
- 所有数据来自真实情境（价格、费用结构），要求准确处理小数和货币单位。
- 核心在于对现实世界定量信息的有效操作。

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3. 能力 #12：执行多步骤程序

（高度相关）

- 解题包含清晰的两个逻辑阶段、三个计算步骤：
  1. 从总支出中扣除会员费；
  2. 用剩余金额除以会员单价，得到租片数量；
  3. 用该数量乘以非会员单价，得到新总支出。
- 各步骤环环相扣，构成一个连贯的多阶段推理与计算流程。

4. 能力 #13：利用对情境的理解来指导或加速数学解题过程

（中等相关）

- 情境明确区分“会员费”和“租赁费”，提示总支出 = 会员费 + 租赁费。
- 若忽略“membership fee is separate”，可能误将 52.50 直接除以 2.50，导致错误。
- 对“会员制度”的常识理解（一次性年费 + 折扣租金）帮助学生正确建模费用结构。

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5. 能力 #1：执行简单计算

（中等相关）

- 所有运算均为基础四则运算，但涉及小数（如 42.50 ÷ 2.50），需一定熟练度。
- 属于支撑性能力，服务于更高阶的策略实施与多步推理。

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6. 能力 #2：选择策略以解决问题

（弱相关）

- 虽然策略未明说，但可行路径基本唯一（先求数量，再算非会员总价）。
- 学生无需在多种方法间权衡（如设方程 vs 算术法），因此策略选择空间有限。